Dataset reference

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Dataset reference

Dataset card: Weather Intelligence Assistant

The assistant's historical weather data covers **8 Indian cities**: Chennai, Delhi, Guwahati, Jaipur, Kolkata, Mumbai, Pune, Shimla. It contains one row per city per day from **2015-01-01 to 2025-01-01**. There is **no data after 2025-01-01**: questions about later dates cannot be answered from this dataset and need a live forecast or a newer data source.

Normal values in this knowledge base are **10-year averages (2015–2024)**, not the 30-year normals that the India Meteorological Department (IMD) uses, so anomalies here may differ from IMD bulletins.

Variables

Column	Meaning	Unit
temp_max	Daily maximum air temperature at 2 m	°C
temp_min	Daily minimum air temperature at 2 m	°C
precip	Daily total precipitation (rain + snow water equivalent)	mm
wind_max	Daily maximum sustained wind speed at 10 m (not gusts)	km/h
humidity	Daily mean relative humidity at 2 m	%
pressure	Daily mean sea-level pressure	hPa
weathercode	Most significant WMO weather code of the day	code (see WMO weather codes document)

Limitations

- The data comes from the Open-Meteo Historical Weather API. It is **reanalysis-based gridded data**, not readings from a single weather station, so local extremes (for example one neighbourhood's rainfall) can be underestimated.
- Days are aggregated in **UTC (GMT)**, not Indian Standard Time; a "day" runs from 05:30 to 05:30 IST.
- Wind is sustained wind at 10 m height. **Gusts and wind at height (e.g. at crane level) are higher** and are not in the data.
- Rainfall is a daily total. **Hourly rainfall intensity and flood water depth are not available.**
- The weather codes in this dataset never include thunderstorm (95–99) or fog (45, 48) codes, so thunderstorms and fog cannot be identified from it.

Event definitions used in this knowledge base

All weather events (heatwaves, cold waves, heavy rain, dry spells, high wind, snowfall) in the event documents are detected automatically from the daily data with the rules below. Thresholds follow India Meteorological Department (IMD) criteria where IMD defines them. "Departure" means the difference from the normal for that calendar date, where the normal is a 15-day smoothed average over 2015–2024.

These are **estimates from gridded data for a single location**. Official IMD declarations require the criteria at several stations in a meteorological subdivision, so an official heatwave may be missing here, or the reverse.

City zones

Zone	Cities
plains	Delhi, Guwahati, Jaipur, Kolkata, Pune
coastal	Chennai, Mumbai
hills	Shimla

Heatwave (at least 2 consecutive days)

- **Plains:** max temperature ≥ 40°C and departure ≥ +4.5°C, or max temperature ≥ 45°C. Severe: max ≥ 40°C and departure ≥ +6.5°C, or max ≥ 47°C.
- **Coastal:** max temperature ≥ 37°C and departure ≥ +4.5°C. Severe: departure ≥ +6.5°C.
- **Hills:** max temperature ≥ 30°C and departure ≥ +4.5°C. Severe: departure ≥ +6.5°C.

Cold wave (at least 2 consecutive days)

- **Plains:** min temperature ≤ 10°C and departure ≤ −4.5°C, or min temperature ≤ 4°C. Severe: min ≤ 10°C and departure ≤ −6.5°C, or min ≤ 2°C.
- **Coastal:** min temperature ≤ 15°C and departure ≤ −4.5°C. Severe: departure ≤ −6.5°C.
- **Hills:** min temperature ≤ 0°C and departure ≤ −4.5°C. Severe: departure ≤ −6.5°C.

Rain

- **Rainy day:** at least 2.5 mm of rain.
- **Heavy rain:** 64.5–115.5 mm in a day. **Very heavy:** 115.6–204.4 mm. **Extremely heavy:** ≥ 204.5 mm. Consecutive heavy-rain days are grouped into one event.
- **Dry spell:** 10 or more consecutive days below 2.5 mm inside the city's main rainy season (June–September for all cities except Chennai, which uses October–December, its northeast monsoon season).

Wind and snow

- **High wind:** daily maximum sustained wind ≥ 40 km/h. Gusts are not in the data and will be stronger.
- **Snowfall:** a day with weather code 71–77 (snow).

Rainy-season onset estimate

The first date on or after May 20 (October 10 for Chennai) from which the next 5 days total at least 30 mm, with at least 3 of those days being rainy days. This is **a rainfall-based estimate, not the official IMD monsoon onset date**, and pre-monsoon showers can make it early.

WMO weather codes used in the dataset

The `weathercode` column uses World Meteorological Organization (WMO) weather interpretation codes. For daily data the code is the most significant weather of that day. Codes in the 50s mean drizzle, 60s mean rain, 70s mean snow, 80s mean showers and 95–99 mean thunderstorms. The last column shows how many city-days in this dataset have each code.

Code	Meaning	City-days in dataset
0	Clear sky	3669
1	Mainly clear	2144
2	Partly cloudy	2007

Code	Meaning	City-days in dataset
3	Overcast	6716
45	Fog	0
48	Depositing rime fog	0
51	Light drizzle	4513
53	Moderate drizzle	2338
55	Dense drizzle	857
56	Light freezing drizzle	0
57	Dense freezing drizzle	0
61	Slight rain	2339
63	Moderate rain	3676
65	Heavy rain	839
66	Light freezing rain	0
67	Heavy freezing rain	0
71	Slight snowfall	30
73	Moderate snowfall	62
75	Heavy snowfall	42
77	Snow grains	0
80	Slight rain showers	0
81	Moderate rain showers	0
82	Violent rain showers	0
85	Slight snow showers	0
86	Heavy snow showers	0
95	Thunderstorm	0
96	Thunderstorm with slight hail	0
99	Thunderstorm with heavy hail	0